

Building An Advanced Navigation System

GPS personal tracking devices and applications are forecast to grow at a rate of 40 per cent—according to ABI Research. Wearable electronics and the electronics of things are the primary contributors to this growth



Car dashboard with GPS panel

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Location information has become integral to our lives due to the increased bearing that it has in the evolution of 'smart services.' It has been implemented in everything from the smart meters in smart grids to mainstream smartphones, motorcycles, cameras, wearable consumer electronics, and all kinds of telematics applications.

Wearable electronics, the *Electronics of Things* and apps designed for machine-to-machine (M2M) communication are significantly driving the adoption of GPS functionality into every device. It is so widespread that spoofed GPS signals can have most dire security implications.

What is it?

In my interactions with junior engineers, I found that everyone considered GPS to be the beginning and the end of positioning technologies. Well, it is not!

GPS belongs to the United States Department of Defence (DoD). It is obvious that the other mighty countries on Earth aren't going to let the USA have all the positioning fun, and that is why we also have the Russian GLONASS, the European GNSS, the Japanese QZSS and our very own Indian IRNSS, amongst others. (For more information on these, please refer the box on next page.)

All these systems are basically a network of very expensive satellites

orbiting our Earth. All of them emit a signal to any navigation system receiver on Earth willing to hear them.

How the system works. The signal transmitted by these satellites is essential to the functioning of the navigation system, due to the electronic ephemeris data that it contains. Receivers on the ground collect this data from the satellites, apply some algorithms to cover for the signal travel time, and finally calculate their location on Earth using trilateration. So, as you can see, receiving the signal intact is of utmost importance to the performance of this system.

Major challenges

As mentioned at the beginning of this article, GPS technology finds a very wide variety of applications. However, there are some problems that are common across the board.

The biggest problem—power consumption. The problem of power consumption is universal, to the point that it feels more like a luxury specification these days. Laptops looked pretty well with their 4-hour battery life until the new range of laptops with 12-hour uptime came in. Now it is annoying when a device battery dies in 'just' four hours.

GPS-enabled devices used to die a quick death when GPS was actually switched on, due to huge power required to keep listening to the satellite and gathering data. Reducing the power consumption not only increases the battery life of existing devices but also makes it more practical to add GPS functionality to new application areas such as wearable electronics and remote sensors where battery life is very important.

How to do it? One way is to go for GPS modules with a reduced update frequency. Higher update frequencies cause the GPS module to gather ephemeris data at a very fast rate and subsequently also push out the NMEA sentences pretty fast. This results in an increased power consumption both by the module and the chip, which has to